CROSS-ASSET QUANT RESEARCH

Miners vs. Gold

GDX vs. Spot Gold

Trade: SHORT GDX / LONG Gold (β-hedged) | **Universe:** Equity ↔ Commodity | **Horizon:** 1–5M | **Conviction:** MEDIUM

Gold has given back most of its August spike, but gold miners kept their extra gain: GDX sits 12% (+2σ) above what gold explains, and Q3 results from Oct 22 should start to close the gap.

SNAPSHOT

Current Gap	**+2.0σ**
Correlation	**+0.85**
Lookback	**18 yrs**
Times This Rich	**4 since ’08**
Long Leg	**Gold (futures)**
Short Leg	**GDX**
Target	**+9% (0.5σ)**
Stop	**−6% (3.0σ)**
R/R	**1.5x**

Executive Summary

RECOMMENDATION: SHORT GDX / LONG GOLD (1 : 1.8) | Gap: +2.0σ | Target: 0.5σ | Stop: 3.0σ | Horizon: 1–5M | **MEDIUM CONVICTION**

01

Economic Mechanism

A miner's profit per ounce is the gold price minus a cost that barely moves (≈ $1,785), so profit swings more than gold: at today's $4,157, a 1% gold move changes profit by P/(P−C) ≈ 1.75%.

The data agree: GDX moves ~1.8x gold (correlation 0.85). When it moves beyond that, something other than gold is driving it.

02

Quantified Dislocation

How far: stripping out gold's effect (GDX ≈ 1.8x gold), GDX sits 12% above its 1-year norm: +2.0σ rich.

How often: higher than on 98% of days since 2008. Only 4 earlier episodes reached +2σ.

Which is wrong: GDX. Gold spiked 15% in Aug, then gave back ~80% as interest rates rose. GDX overshot on the way up (+42% vs. ~27% expected), then fell only in line with gold, so it still holds that ~12% extra. Profit forecasts never rose (now −2%).

03

Trade & Failure Conditions

Trade: Short $1 GDX, buy $1.8 of gold, so gold's direction cancels out.

Convergence: GDX gives back its extra (fair ≈ $77.8) or lags if gold rises.

Trigger: Q3 results. Gold's drop is public but miners' costs aren't, and they are likely up: diesel +28% since June, and Kinross has already raised its cost guidance.

Permanent break: miners stop being pure gold bets (e.g. shifting to copper) or costs rise as fast as gold (royalties, taxes). Warning sign: correlation below 0.6.

Relationship & Economic Rationale

Economic Mechanism

Miners sell gold at the market price. Their costs change slowly, so profit swings more than gold: gold $4,000 → $4,400 (+10%) lifts profit per ounce $2,215 → $2,615 (+18%). Today that multiplier is gold ÷ (gold − cost) ≈ 1.75x, close to the 1.8x measured from the past year of weekly prices.

Transmission Channels

1. **Profit margin (+, main)**: gold ↑ → profit per ounce ↑ ~1.75x (costs barely move) → GDX ↑.
2. **Investor buying (+, can overshoot)**: gold ↑ → stock investors buy miners as an amplified gold bet → GDX ↑, sometimes faster than profits justify.
3. **Cost pushback (−, slow)**: gold ↑ → higher royalties, taxes and wages → costs ↑ → GDX trailed 1.8x gold in 15 of 19 years (~8%/yr over 5 yrs).

Why the Link Should Persist

Gold is most of a miner's revenue and most costs are fixed in the short run, so the link is built in: cost math predicts 1.75x and the data shows 1.8x. Correlation held at 0.78–0.89 in every period tested. It breaks only if gold stops being most of revenue (locking in gold prices in advance, shifting to copper) or costs start rising with gold (royalties and taxes tied to the gold price). Neither is happening: correlation 0.85, cost math still matches.

Key Numbers

Asset Y	**GDX + dividends**
Asset X	**Spot gold (XAU)**
Data Range	**2006–2026**
Data Source	**Bloomberg**
Measure Used	**GDX − β × gold**
Correlation (Full)	**+0.81**
Correlation (1Y)	**+0.85**
Avg Gap (typical |z|)	**1.2σ**
Current Gap (vs. 1Y avg)	**+12.1%**
Current Z-Score	**+2.0σ**
Largest Past Gap (rich / cheap)	**+2.7σ / −5.7σ**

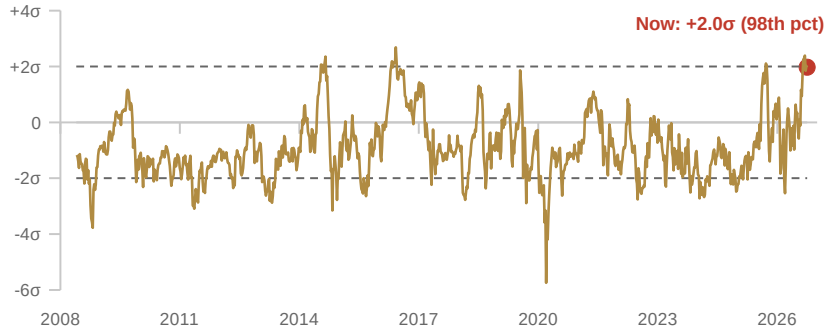
Historical Evidence

How We Measured It

Bloomberg daily prices, 2006–2026. Each week, GDX's sensitivity to gold is re-measured from the past year (now 1.8). Track long $1 GDX / short $1.8 gold against its own past year.

Why this measure: a plain GDX/gold ratio rises whenever gold does, so it would flag a gap even when miners act normally. Re-measuring weekly keeps the hedge right as sensitivity shifts (1.4x in 2012, 3.2x in March 2020). For today a simple fixed 1.8x spread also agrees: +1.9σ.

GDX − β × Gold, Z-Score (2008–2026)



Does the Relationship Hold Over Time?

Period	Corr	Avg Gap	Max Gap	# ±2σ	Held?
Full (2008–26)	+0.81	1.2σ	−5.7σ	21	Y
Last 10Y	+0.82	1.2σ	−5.7σ	14	Y
Last 5Y	+0.83	1.2σ	−3.3σ	8	Y
Last 3Y	+0.83	1.3σ	−2.9σ	5	Y
Last 1Y	+0.85	0.8σ	+2.6σ	3	Y
COVID crash ’20	+0.89	2.9σ	−5.7σ	1	Partly
GFC crash ’08	+0.78	2.2σ	−4.1σ	1	Partly

Link held in every period tested (correlation 0.78–0.89). Every gap past 3σ was on the cheap side, the largest in crashes (Covid selling: −5.7σ in 2020). The rich side is calmer: never past +2.7σ, and all 4 earlier +2σ episodes fell back to +0.5σ within 5 months.

Current Dislocation — Why Now?

WHY IS THE RELATIONSHIP DISLOCATED NOW?

Size of the Deviation

+2.0σ, or 12% above its 1-year average: higher than on 98% of days since 2008. With the gap back at its 1-year average, GDX would be ≈ $77.8, not $87.8 (−11%).

What Caused It

Gold spiked 15% in August in part due to a Treasury bond move, then gave back ~80% as interest rates rose. Additionally fueled by record Q2 cash flow, miners overreacted on the way up (+42% vs. ~27% expected) but fell only in line with gold, keeping the extra.

Which Market Is Right?

Gold. The 10-year yield rose from 4.6% to 5.3% (highest since 2007), which by the past year's pattern explains about all of gold's −11%. GDX oddly kept its ~12% extra even as miners' outlook got worse: profit forecasts −2%, diesel +28% since June.

Asymmetric Setup

+9% if the gap closes to +0.5σ, −6% at the +3σ stop: 40% hit rate breaks even; past stretched entries hit 64% (only ~5 episodes). Why now: gold has already given back its spike. Q3 (from Oct 22) will bring new miner numbers.

NORMALIZATION CATALYSTS

0–1 Months — **Q3 results: the cost reveal**

Kinross preview (Sep 23: −15% vs GDX −6%). Newmont Oct 22, Agnico and Kinross Oct 28, Barrick Nov 9: first cost data since the spike.

Expected: ~+1.0σ if costs disappoint (~+1.7σ if they don't)

1–3 Months — **Steady drift back**

No single event needed: past rich gaps closed gradually, with miners slipping little by little most days. Median ~6 weeks to target; 3 of 4 earlier +2σ cases reached it within 3 months.

Expected: ~+0.5σ, our target (take profit here)

4–5 months 2027 — **2027 outlooks: cost reset**

Barrick (~Feb 4) and Newmont (~Feb 18) set 2027 cost and output targets: the first full-year budgets at today's ~$4,200 gold (Newmont's 2026 plan assumed $4,500) and today's costs (diesel +28% since June).

Our backstop if the gap stalls; hard exit by end-Feb

Market vs. Normal Relationship

What the Market Says vs. What the Relationship Says

Metric	Today	If Normal	Gap	Why It Should Close
GDX price	$87.80	$77.82	−11.4%	Where GDX would be with the gap back at its 1-year average
Gap (z-score)	+2.0σ	0.0σ	−2.0σ	vs. its 1-year average; higher than on 98% of days since 2008
What gold says	Rally over	—	—	Gold gave back ~80% of its August spike as interest rates rose.
What GDX says	Rally still on	—	—	GDX kept all of its extra gain over gold, priced as if gold were ~$4,440 (+7%). This is the mismatch.
Correlation (1Y vs. full)	+0.85	+0.81	−0.04	The link is not weakening

Source of Edge

Why the Gap Exists

Miners overreacted to a 15% gold spike: two-thirds of their extra came in early August, right after record Q2 cash flow reports; a Treasury bond-buying move on Aug 19 then pushed gold, and miners, higher still. GDX moved ~2.8x gold vs. its usual ~1.8x. Gold is now a net +2.7%; GDX is still +18.5% (normal: ~+5%).

Why It Hasn't Closed

Miners trade on their own reported numbers, and most have not come out since the spike. The one exception: Kinross cut its outlook Sep 23 and fell 15% in two days vs. GDX −6%. Nothing forces the gap shut (you can't swap a miner share for gold), so it waits for Q3 reports.

What Proves You Right

Q3 reports show costs up or outlooks cut, and GDX does worse than its usual 1.8x gold ratio implies, while the link holds (correlation stays above 0.6). Wrong if Q3 shows lower costs, or lifts profit forecasts enough to explain the 12% gap beyond gold.

Trade Expression

POSITION PARAMETERS

Structure	**SHORT GDX / LONG Gold**
Instruments	**GDX ETF / micro gold futures**
Sizing	**$1.82 gold per $1 GDX**
Entry	**+2.0σ (GDX $87.80)**
Target	**+9.2% fixed (gap at +0.5σ)**
Risk/Reward	**1.5x (break-even 40%)**
Stop	**−6.1% fixed (gap at +3.0σ)**
Horizon	**Through Feb 2027 (~5 months)**
Exit Plan	**Take profit at +9.2%; exit by end-Feb**

TRADE RATIONALE

Why This Expression

Shorting GDX alone would mostly be a bet that gold falls. Adding $1.8 of gold per $1 of GDX cancels gold's direction. We only keep the miners' excess move.

Catalyst

Q3 results (Oct 22–Nov 9) reveal costs; gaps then drift back gradually; 2027 outlooks (Feb) reset budgets at today's gold and costs.

What Convergence Looks Like

GDX lags gold rather than crashing: miners keep moving ~1.8x gold but slip on their own news (cost misses, outlook cuts). Past gaps took ~6 weeks to reach target.

Cost of Holding

≈3–4%/yr (~1.5% over 5 months): gold futures cost about the interest rate (~4%) on $1.82, partly offset by interest on the $1 GDX short, plus a small GDX borrow fee (0.43%).

Liquidity

GDX: ~$26bn fund, easy to borrow and trade. Micro gold futures = 10 oz (≈ $41.7k): a $100k GDX short pairs with ~4.4 contracts.

Thesis Invalidation

Exit if: (1) loss hits 6%; (2) 1-year correlation drops below 0.6; (3) Q3 shows miners earned the jump: costs fall or profit forecasts rise enough to explain the 12% gap.

Failure Conditions — When Are We Wrong?

Risk	Sev	Prob	Description	Mitigant / Hedge
Miners Earned It	HIGH	MED	Q3 shows the extra was deserved: Newmont and Agnico hold costs as lower royalties offset diesel, raise payouts, and buyers keep paying takeover premiums. Then this is a lasting re-rating, not a chase.	Exit if Q3 shows lower costs, or lifts profit forecasts enough to explain the 12% gap.
Gold Chase Returns	MED	MED	Gold surges again and miners chase it at 2.8x, as in August. Triggers: hopes the Fed stops hiking after weak September jobs, or Treasury expanding its bond buying again on Nov 4.	Fixed −6% stop (gap at +3σ); don't add on rallies. If gold closes above its August high ($4,657), halve the position: the August story is back.
Stock Market Rally	MED	MED	Our hedge covers gold, not stocks. Miners also trade as stocks, so the gap rises ~0.3–0.6% for every 1% S&P gain. A 10% rally (~30% of 5-month windows since 2008) would cost ~3–6%, most of the room to our stop.	Accept it as a known side-bet, or cancel it: buy ~$0.4 of S&P futures per $1 of GDX.
Link Breaks	HIGH	LOW	Miners stop tracking gold: copper becomes a bigger share of Newmont's and Barrick's profits, or more costs get tied to the gold price, so 1.8x is the wrong hedge. Never seen: the 1-year correlation hasn't dipped below 0.67 since 2007 (now 0.85).	Re-check the ratio weekly and the cost math (gold ÷ (gold − cost) ≈ 1.75x) each quarter. Exit if the 1-year correlation falls below 0.6; its biggest 3-month drop ever was 0.16, so that would be a real break.
Divergence Widens	MED	MED	The gap widens before it closes: 2016 ran from +1.5σ to +2.7σ; an Aug 24 entry this year at +1.5σ was down 6.9% by Sep 8, past a 6% stop. About 1 in 3 past stretched entries hit the stop.	Fixed −6% stop (gap at +3σ) and a hard exit by end-Feb.